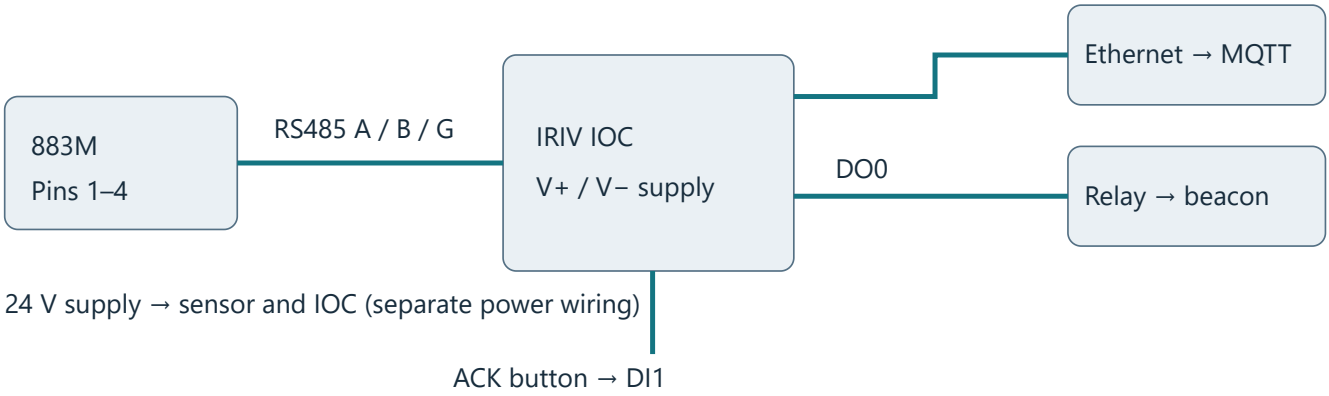
Wilcoxon 883M + IRIV IOC

Single-sensor demonstration · Wiring, parts and commissioning guide

> Use with the equipment manuals and the selected firmware. The film is illustrative. The current custom gateway reads one configured sensor ID; this guide does not enable a 32-sensor network. Isolate power before wiring.

Signal and power overview



Signal paths only. Use the terminal table below for supply returns, commons and coil wiring.

Connection table

From	To	Notes
24 V PSU +	IOC V+, 883M pin 1 (power)	IOC takes 10–30 V DC, the 883M 6–24 V DC. Keep the rail at 24.0 V, not above.
24 V PSU −	IOC V−, 883M pin 2 (ground)	Common 0 V.
883M pin 3 (signal +)	IOC RS485 A	Use 4-conductor shielded cable with an M8 4-pin plug, keeping signal + and − on one twisted pair.
883M pin 4 (signal −)	IOC RS485 B	If there are no replies, swap A and B first.
883M ground (0 V)	IOC RS485 G	Gives the isolated port its reference. Termination switch: 120 Ω ON (the IOC ends the bus).
Cable shield	Cabinet earth, one end only	The 883M ties the shield to its housing (pin 5).
24 V +	IOC DO0 (first terminal)	DO0 is a solid-state output on its own pair of terminals, rated 50 V / 500 mA.
IOC DO0 (second terminal)	Relay coil A1; A2 to 0 V	Use a 24 V DC relay and fit a flyback diode across the coil.
Relay NO contact	Beacon + buzzer	The relay contact carries the beacon's current, not the IOC output.
24 V + → push button (NO)	IOC DI1	Wire S/S0..4 to 0 V so DI1 sees +24 V when pressed. A rising edge acknowledges.

From	To	Notes
Optional button	IOC DI0	A rising edge starts an X/Y/Z waveform capture.
IOC Ethernet	Site network → MQTT broker → Viking Analytics	Plain MQTT on port 1883. See the TLS note below.

Parts list

Qty	Item	Select for the installation
1	Wilcoxon 883M	Mounting adapter suitable for the machine
1	IRIV IO Controller	Compatible custom 883M gateway firmware
1	Regulated 24 V DC supply	Size for controller, sensor, relay and alarm loads
1 run	Shielded M8 4-pin cable	Verify pinout; RS485 pair, environmental rating and bend radius
1 each	24 V relay, socket and coil suppression	Contacts rated for the selected beacon/buzzer load
1 each	Beacon/buzzer and normally-open ACK button	24 V devices suitable for the environment
As needed	Enclosure, DIN rail, fuse protection, terminal blocks, glands, clamps and Ethernet cable	Size and rate for the final cabinet and cable route
As needed	RS485 termination	At both physical bus ends; avoid duplicate termination

Commissioning checklist

☐ Confirm sensor model, firmware build and all terminal labels.

☐ Mount sensor securely; identify axes; support cable clear of rotating parts.

☐ With power isolated, check continuity, polarity, commons, shield and termination.

☐ Confirm supply voltage at the sensor and IOC under load.

☐ Match baud, parity, stop bits and the configured sensor ID.

☐ Check fresh readings and sensor identity in the gateway dashboard.

☐ Configure broker, topic and credentials; verify readings at the receiver.

☐ Choose machine-specific limits, delay and hysteresis.

☐ Test alarm, acknowledgement while still high, return-to-normal and sensor-loss behavior.

☐ Record baseline, operating conditions, configuration and test results.

Reference documents

Wilcoxon 883M specifications and setup manual
Custom gateway firmware and wiring documentation
Cytron RS485 documentation

Demonstration guide · review against the actual equipment before commissioning.